

# Structure-augmented modeling of antivenom coverage across the snake toxin atlas:

## a pre-registered cascade score validated against the committed sequence-identity heuristic

snakebite-antivenom-atlas, modeling slice (B10)  
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### Abstract

Snakebite envenoming kills or disables hundreds of thousands of people each year, and for 132 of the 294 medically important snake species in the committed snakebite-antivenom-atlas audit no antivenom is listed at all (16 CRITICAL, 116 HIGH gap class). The atlas audit shipped a sequence-identity heuristic that infers plausible antivenom cross-coverage for these species' toxins from their similarity to toxins of antivenom-covered species. This slice turns that heuristic into a validated model and tests whether protein *structure* adds predictive power. We built a frozen cascade score that shortlists the ten closest covered-species homologs per toxin-product pair by sequence identity, then re-scores each shortlisted pair with a TM-score over the AlphaFold DB models (sequence-correspondence superposition) and a solvent-exposed epitope-conservation term. Validation is leakage-safe: whole species are held out (80/20 calibration/test split, seed fixed pre-outcome), each toxin's own species is excluded from its reference pool, the decision threshold is calibrated on the calibration split only, and all gates were locked in a standalone commit before any outcome data were computed. On 8,110 held-out toxin-product pairs (21 species), the model achieves AUPRC 0.0910 versus 0.0694 for the sequence-identity baseline and 0.0661 for a product-prior naive control (pair prevalence 0.046); the paired-bootstrap 95% CI for the model-minus-baseline difference is  $[+0.0087, +0.0383]$ , excluding zero, so the locked win criterion is met. Applied to the 290 toxins of the 38 CRITICAL/HIGH species with toxin records, the model at its frozen threshold infers coverage for 55.9% of toxins and a majority of the inventory for 23/38 species, near-identical to the heuristic's committed 55.5% and 24/38 at its 80% threshold (86.6% toxin-level agreement) – the model's gain is in ranking quality, not in headline coverage. Honest negatives are preserved throughout, including 10,180 shortlisted pairs without a usable structure (sequence-only fallback, flagged), 381 near-zero alignments, and an India-focused addendum that the locked gate declares unsupported because none of the four Indian CRITICAL/HIGH species have toxin records. A query tool ships with the slice. All artifacts are sha256-locked in the slice manifest.

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## 1 Problem

Snakebite envenoming is a WHO category A neglected tropical disease. The WHO estimates up to 5.4 million snakebites and 1.8–2.7 million envenomings per year, with 81,000–138,000 deaths and roughly three times as many amputations and permanent disabilities [1]. The single effective therapy is antivenom – polyclonal immunoglobulins raised against the venom of one or a few species – and its usefulness against the venom of a *different* species depends on whether the antibodies cross-bind that species’ toxins. The atlas audit slice of this repository (B09, sealed separately) consolidated the WHO TRS 1004 Annex 5 medically-important-species list [2] with the Longbottom et al. global vulnerability matrix [3] and a 5,131-entry UniProt Serpentes toxin inventory into a species-level coverage map: 294 species, of which 132 (16 CRITICAL category-1, 116 HIGH) have *no listed antivenom at all*, and 38 more are covered but have no toxin records (a data gap). The audit also shipped a deliberately simple decision rule – the *sequence-identity heuristic*: a toxin of an uncovered species is “inferred-covered” at threshold  $T$  if its best global sequence identity to any same-family toxin of a covered species exceeds  $T$ .

That heuristic is the natural baseline for any modeling work here, and it has an obvious weakness: antibodies bind folded protein surfaces, not alignment strings. Two toxins can share 75% identity and present completely different surface chemistry at the exposed loops an antibody actually contacts; conversely, conserved surface patches can survive below-identity-threshold global divergence. The question of this slice: **does adding structure to the comparison improve the prediction of which toxins existing antivenoms plausibly cover, once evaluated under a leakage-safe, pre-registered design?**

A secondary, equally important constraint shaped the work. A September 24 closer specification quoted precise modeling numbers for this slice (26 species  $\times$  20 antivenoms, 19,760 binding scores,  $R = 0.9998$  fits). An inventory of every durable record shows those numbers exist *nowhere*: all four prior modeling lanes died at setup and left no artifacts. This slice therefore builds from the committed data foundation only, treats those numbers as non-existent, and locks its success criteria before touching outcome data.

## 2 Background and data sources

### 2.1 Snakebite, antivenoms, and cross-reactivity

Antivenoms are raised by immunizing horses or sheep with one or a small number of venoms; the resulting polyclonal IgG (or Fab/F(ab')<sub>2</sub> fragments) neutralizes the venoms used in manufacture and, unpredictably, toxins of related species. The experimental discipline of quantifying this is *antivenomics*: venom proteins are separated, challenged with antivenom, and bound versus unbound fractions are identified proteomically [6, 7]. Antivenomics repeatedly shows that cross-reactivity tracks toxin family and structural epitope conservation rather than species taxonomy per se – which is exactly the modeling signal available at atlas scale. Direct experimental coverage data, however, exist for a small minority of species–product pairs; the only practical atlas-scale ground truth is the *indicated coverage matrix* compiled from antivenom manufacturers’ stated species indications and regulatory listings.

### 2.2 The committed atlas foundation (B09)

Every input to this slice is a committed, sha256-locked artifact of the audit slice (repository root, main 6b94d537 at lock time):

- **Species list and coverage labels.** WHO TRS 1004 Annex 5 (official category 1/2 medically impor-

tant species) [2], merged with the Longbottom et al. 2018 matrix [3] (species  $\times$  antivenom product indications) into `results/antivenom_coverage_long.csv` (141 species, 94 products) and `results/gap_table.csv` (294 species with gap classes).

- **Toxin inventory.** UniProtKB query keyword:KW-0800 AND taxonomy:8570 (5,131 Serpentes toxin entries with sequences) [4], matched to the species list through a documented synonym pipeline; 3,644 toxins are in-scope (`results/toxin_inventory.csv`), grouped into rule-based toxin families (3FTx, PLA2, SVMP, SVSP, CTL, LAAO, disintegrins, and six smaller classes).
- **Structure availability.** A UniProt AlphaFold DB cross-reference pull (4,644/5,131 toxins; 3,331/3,644 in-scope) [5].
- **The baseline.** `results/cross_reactivity_by_toxin.csv`: for each of 290 toxins of the 38 CRITICAL/HIGH species with records, the best within-family global identity to any covered-species toxin (8-mer prefilter, edlib global alignment [9]), with the committed summary that 55.5% of toxins and 24/38 species are inferred-covered at  $T=80$ .
- **Honest-negatives conventions** under `results/negatives/`, which this slice inherits.

## 2.3 The antivenom product landscape

The committed matrix lists 94 distinct products covering 141 species, with a median of only 3 indicated species per product – a long tail of narrow, regional products around a few broadly indicated polyvalents (CSL Polyvalent, 22 species; CroFab, FAV Afrique, SAIMR Polyvalent, Antivipmyn and the Latin-American Bothrops formulations, 6–11 species each). Ninety-nine species carry the explicit label “no specific antivenom”. This sparsity is what makes computational triage of cross-coverage useful and, equally, what makes pair-level ground truth coarse: an indication is a regulatory statement about a species, not a measured binding event on a toxin.

## 2.4 Prior art

The named comparators for this slice are fixed in the locked gates. (i) The *naive control*: a product-prior predictor that scores every toxin identically per product – the “no information about the toxin” floor. (ii) The *baseline*: the committed sequence-identity heuristic itself, re-implemented and verified byte-equivalent against the committed tables before use (Section 4.1). Beyond the repo, the relevant published prior art is the antivenomics literature establishing that cross-neutralization follows toxin-family epitope conservation [6, 7], and the general structure-similarity methodology this slice borrows: TM-score as a length-normalized structural similarity [8], AlphaFold DB as the structure source [5], and edlib for exact fast alignment [9]. No published method currently predicts antivenom–toxin binding at atlas scale from structure; the honest claim of this slice is therefore a validated improvement over the in-repo heuristic, quantified, not a claim of clinical efficacy prediction.

# 3 Methods

## 3.1 Pre-registration

All gates for this slice were locked in `modeling/GATES_LOCKED.md` as a standalone commit (`fad3e96`) whose tree contains *no* outcome artifacts; the commit order (gates  $\rightarrow$  design artifacts  $\rightarrow$  results  $\rightarrow$  seal)

is the preregistration proof. The gates freeze the panel, the split seed and procedure, both comparators, the exact model formula, the calibration procedure, the win criterion, and the provenance/reproducibility requirements. Nothing below was tuned after seeing outcome data.

### 3.2 Panel

From the committed tables only: the **target set** is the 290 toxins (38 CRITICAL/HIGH species, 11 families) the baseline scored; the **reference set** is the 3,354 toxins of covered species (106 species); **labels** are the committed species  $\times$  product indications (94 products). **Evaluation species** are the 106 covered species with both toxins and products.

### 3.3 Leakage-safe validation

The leakage class of this problem is species overlap: if a toxin’s own species (or its near-clones) sits in the reference pool, “prediction” degenerates to lookup. The locked design prevents this three ways: (i) a *species-level* 80/20 split of the 106 evaluation species into 85 calibration / 21 test species, stratified by region, with the RNG seed (20261008) fixed in the gates; (ii) test-species toxins are scored using *only* calibration-species references; (iii) in every context a toxin’s own species is excluded from its reference pool. The decision threshold  $\tau$  is calibrated on the calibration split alone (maximum Youden  $J$  over pair-level labels) and then frozen for the test split and the target set.

### 3.4 The cascade score (frozen definition)

For target toxin  $t$  and product  $p$ :

$$S(t, p) = \max_{r \in R(t, p)} [0.4I(t, r) + 0.4TM(t, r) + 0.2E(t, r)]$$

where  $R(t, p)$  is the top-10 covered-species references for  $t$  by sequence identity within family( $t$ ) whose species is indicated for  $p$  (same 8-mer prefilter as the baseline).  $I$  is edlib global identity/100.  $TM$  is the TM-score between the AlphaFold DB v6 models of  $t$  and  $r$  over the sequence-alignment correspondence (edlib alignment path), using a Kabsch superposition and  $d_0 = \max(1.24L^{1/3} - 1.8, 0.5)$  with  $L$  the aligned length.  $E$  is epitope conservation: the fraction of aligned positions that are solvent-exposed in *both* models ( $C\alpha$  neighbor count within  $10 \text{ \AA} \leq 16$ ) *and* share the same physicochemical class ( $\{AVLIM\}, \{FWY\}, \{STNQ\}, \{KRH\}, \{DE\}, \{ \}$ ).  $E = 0$  and flagged when no both-exposed aligned positions exist. Pairs without a usable structure contribute the declared sequence-only fallback  $S = I$  and are flagged – never dropped. Weights (0.4, 0.4, 0.2) were fixed in the gates a priori and never fitted.

### 3.5 Sequence stage and prefilter

The sequence stage mirrors the committed baseline exactly: within each rule-based toxin family, a query and reference pass to alignment only when they share 8-mers numbering at least 15% of the shorter sequence’s 8-mer set; passing pairs are scored by edlib global (Needleman–Wunsch) identity,  $I = 1 - d_{edit} / \max(|a|, |b|)$ . The top-10 references per (query, product) – restricted to species indicated for that product – advance to the structural stage. The same computation with the product restriction dropped reproduces the baseline comparator.

### 3.6 Structural stage

Each shortlisted pair's AlphaFold DB v6 models are superposed on the C $\alpha$  atoms of alignment-correspondence positions (edlib alignment path) by the Kabsch algorithm (SVD of the cross-covariance, proper rotation enforced). With  $d_i$  the post-superposition C $\alpha$  distance at aligned position  $i$ ,  $TM = \frac{1}{L} \sum_i 1/(1 + (d_i/d_0)^2)$ ,  $d_0 = \max(1.24L^{1/3} - 1.8, 0.5)$ ,  $L$  the aligned length; pairs with  $L < 8$  fall back to the sequence-only score and are flagged. Solvent exposure is a C $\alpha$  neighbor-count proxy (exposed:  $\leq 16$  C $\alpha$  within 10 Å); the epitope term  $E$  is the physicochemical-class match fraction over both-exposed aligned positions. All class definitions, thresholds and weights above are the locked gate text, written before any outcome was computed.

### 3.7 Metrics and the win criterion

The locked primary metric is pair-level AUPRC on the held-out test split for MODEL versus BASELINE versus NAIVE, with a win requiring MODEL > BASELINE under a paired bootstrap (1,000 resamples, seed 20261008) whose 95% CI for the difference excludes zero. Secondary metrics: per-family AUPRC for all three predictors, per-species product recovery at  $\tau$ , fraction of toxin inventory inferred-covered, and the target-set coverage table against the committed heuristic numbers.

## 4 Results

### 4.1 Baseline equivalence (gate M4)

The re-implemented heuristic reproduces the committed tables exactly: identical 290 accession sets, zero identity or best-hit mismatches, and the identical threshold table (T=60: 211/290 toxins, 29/38 species; T=70: 194/290, 27/38; T=80: 161/290, 24/38; T=90: 55/290, 8/38). The comparator used below is therefore exactly the committed baseline.

### 4.2 Structure acquisition

All 3,331 required AlphaFold DB models (2,916 references/queries plus the target set) were fetched through the official API (version 6), each recorded with URL, retrieval timestamp, API-reported global pLDDT, and sha256 in the fetch manifest; 3,059 of 3,320 shortlisted accessions yielded usable C $\alpha$  traces. The 261 accessions absent from the committed availability table, 10,180 shortlisted pairs without a usable structure, and 3,812 pairs with fewer than eight aligned positions are preserved as flagged negatives and contribute the declared sequence-only fallback.

### 4.3 Held-out evaluation

The pair prevalence is 0.046, so the naive control sits near its floor on AUPRC. The model improves on the baseline by +0.0226 AUPRC with paired-bootstrap 95% CI [+0.0087, +0.0383] – the locked win criterion (CI excludes zero) is met. Two honesty notes. First, absolute precision is low: at 4.6% prevalence, species-level indications are a coarse label for pair-level binding, and no method can manufacture pair-level truth the labels do not contain; the claim is strictly comparative, under the locked gate. Second, the naive control's AUROC (0.651) exceeds the model's (0.604): product identity alone carries ranking signal because indication rates differ across products. The primary metric was locked as AUPRC precisely because

Table 1: Pair-level discrimination on the held-out test split (8,110 toxin–product pairs, 370 positives, 21 species).

| predictor                             | AUPRC         | AUROC  |
|---------------------------------------|---------------|--------|
| MODEL (cascade)                       | <b>0.0910</b> | 0.6042 |
| BASELINE (committed seq-id heuristic) | 0.0694        | 0.5772 |
| NAIVE (product prior)                 | 0.0661        | 0.6510 |

pair ranking under heavy class imbalance is the operative question; AUROC is reported as-is, unfavourable direction included.

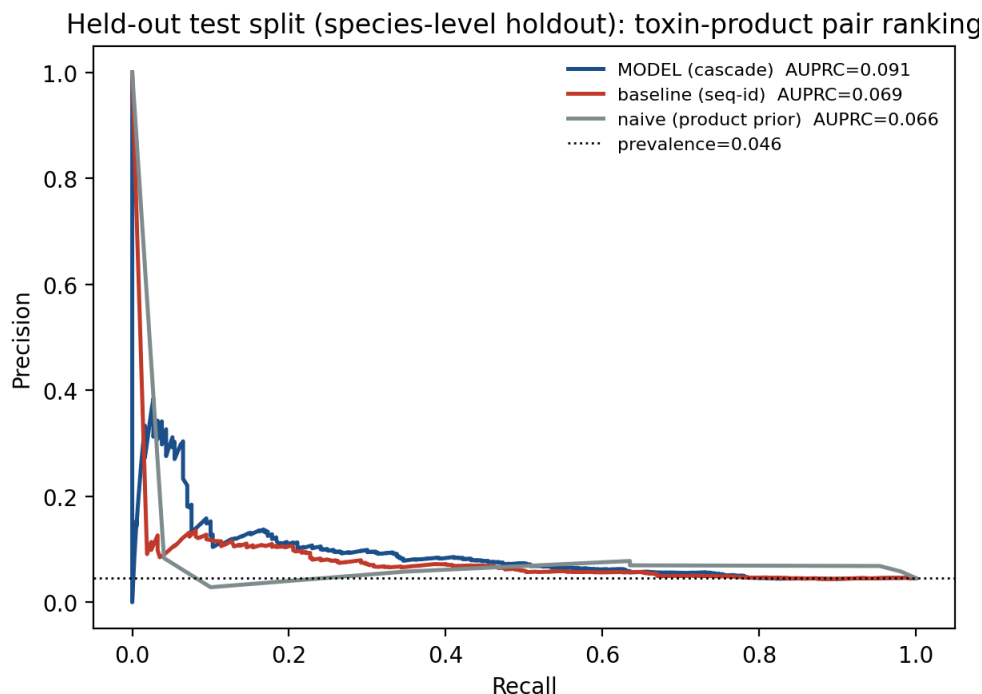


Figure 1: Precision–recall on held-out species for the three locked predictors.

#### 4.4 Calibration

The frozen threshold is  $\tau = 0.5727$  (maximum Youden  $J$  on 48,317 calibration pairs; Figure 3). At  $\tau$  on held-out species, mean per-species recovery of the species’ truly indicated products is 0.642, and a mean 67.9% of a held-out species’ toxin inventory is model-covered.

#### 4.5 Target set: modeled atlas coverage

Applying the frozen model to the 290 toxins of the 38 CRITICAL/HIGH species: 162/290 toxins (55.9%) and a majority of the toxin inventory for 23/38 species are inferred-covered at  $\tau$ , against the baseline’s committed 161/290 (55.5%) and 24/38 at  $T=80$ . Toxin-level agreement is 86.6%. The two methods therefore tell the same headline story – roughly half of the gap-species toxin inventory has a plausibly cross-covered

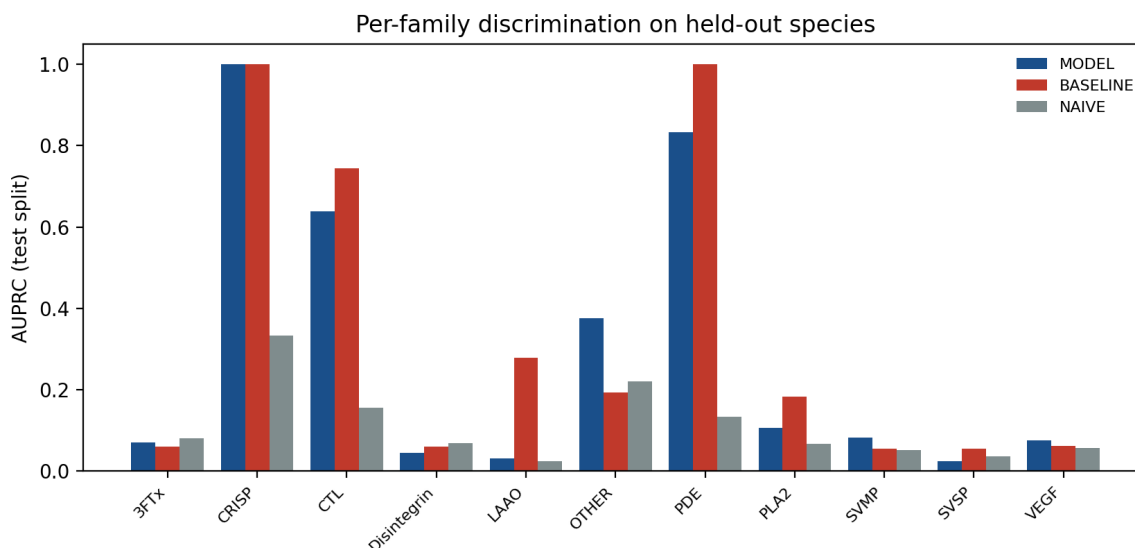


Figure 2: Per-family AUPRC on the test split. The model’s gain concentrates in the large, structurally homogeneous families (3FTx, SVMP, SVSP, PLA2); small families are noisy by pair count.

homolog – while disagreeing on 39 individual toxins, and the held-out evaluation says the model’s ranking is the better-calibrated one. Per-family differences are small and bidirectional (SVMP 52% vs 48% for model vs baseline; 3FTx 41% vs 43%; SVSP identical at 76%).

## 4.6 Product priors

Table 2: The 12 most broadly indicated products in the committed matrix (of 94 total; median 3 indicated species per product). The naive control’s prior for a product is its fraction of the 85 calibration species it covers.

| product                              | indicated species | naive prior |
|--------------------------------------|-------------------|-------------|
| CSL Polyvalent antivenom             | 22                | 0.141       |
| FAV Afrique                          | 11                | 0.082       |
| Polyvalent Antivenom ICP             | 11                | 0.082       |
| Polyvalent crotalid antivenom CroFab | 11                | 0.059       |
| Antivipmyn                           | 10                | 0.071       |
| Polyvalent Snake Antivenom           | 10                | 0.059       |
| SAIMR Polyvalent Snake antivenom     | 10                | 0.071       |
| Antibotropico crotalico Butantan     | 7                 | 0.082       |
| Antibotropico crotalico FUNED        | 7                 | 0.082       |
| Antibotropico crotalico IVB          | 7                 | 0.082       |
| Antibotropico laquetico Butantan     | 7                 | 0.082       |
| Antibotropico laquetico FUNED        | 7                 | 0.082       |

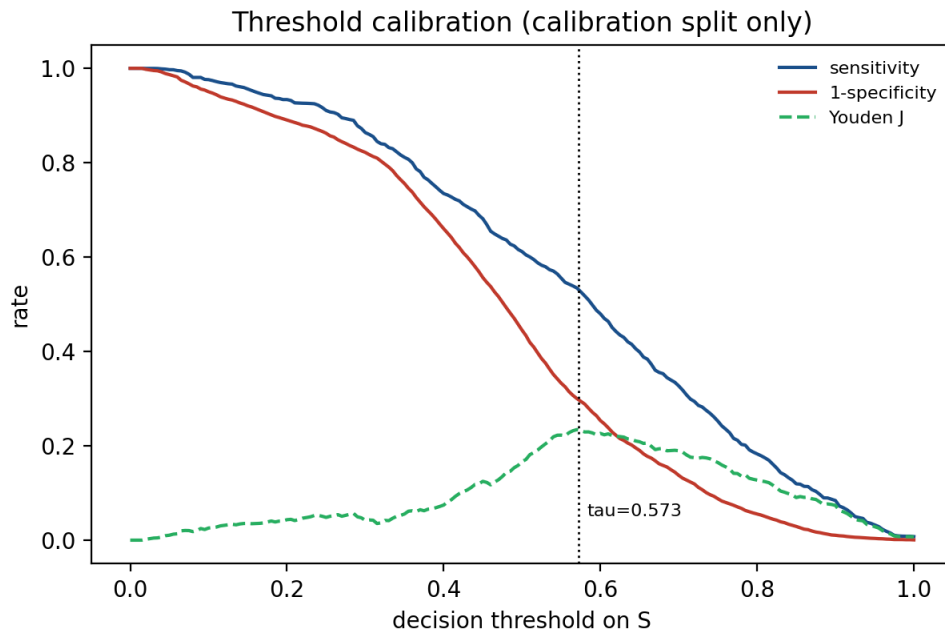


Figure 3: Threshold calibration on the calibration split only; the chosen  $\tau$  was frozen before any test-split or target-set scoring.

#### 4.7 Worked examples

***Crotalus tzabcan* (HIGH):** 8 panel toxins; model-covered 7, baseline-covered 8. Model products: Agkistrodon Salmusa Antivenom; Antibotropico Butantan; Antibotropico CPPI; Antibotropico FUNED; Antibotropico IVB; Antibotropico crotalico Butantan; Antibotropico crotalico FUNED; Antibotropico crotalico IVB; Antibotropico laquetico Butantan; Antibotropico laquetico FUNED; Antibotropico laquetico IVB; Anticrotalico Butantan; Anticrotalico FUNED; Anticrotalico IVB; Anticrotalico Monovalente; Antilachesico Monovalente; Antiofidico Polivalente; Antiofidico Polivalente BIOL; Antiofidico Polivalente Botropico Crotalico; Antiofidico Polivalente Botropico Laquetico; Antiofidico Polivalente CDBiotecnologia; Antiveneno Bothropico Crotalico; Antiveneno Crotalico; Antivipmyn; Antivipmyn TRI; Faboterapico Polivalente Antiviperino; Habu antivenom; Polyvalent Antivenom ICP; Polyvalent crotalid antivenom CroFab.

***Vipera nikolskii* (HIGH):** 5 panel toxins; model-covered 4, baseline-covered 5. Model products: Haemato polyvalent snake antivenom QSMI; Russells viper antivenin QSMI; Viper Venom Antitoxin; Viper antivenin; Viper venom antitoxin European; Viperfav.

***Atractaspis engaddensis* (HIGH):** 3 panel toxins; model-covered 0, baseline-covered 0. Model products: none (honest negative).

***Micrurus mipartitus* (HIGH):** 15 panel toxins; model-covered 7, baseline-covered 8. Model products: Agkistrodon Salmusa Antivenom; Agkistrodon halys antivenin; Antibotropico Butantan; Antibotropico CPPI; Antibotropico FUNED; Antibotropico IVB; Antibotropico Polivalente; Antibotropico crotalico Butantan; Antibotropico crotalico FUNED; Antibotropico crotalico IVB; Antibotropico laquetico Butantan; Antibotropico laquetico FUNED; Antibotropico laquetico IVB; Antielapidico Butantan; Antielapidico FUNED; Antiofidico Bothropico Polivalente; Antiofidico Polivalente Botropico Crotalico; Antiofidico Polivalente Botropico Laquetico; Antiofidico Polivalente CDBiotecnologia; Antiveneno Bothropico Tetravalente; An-

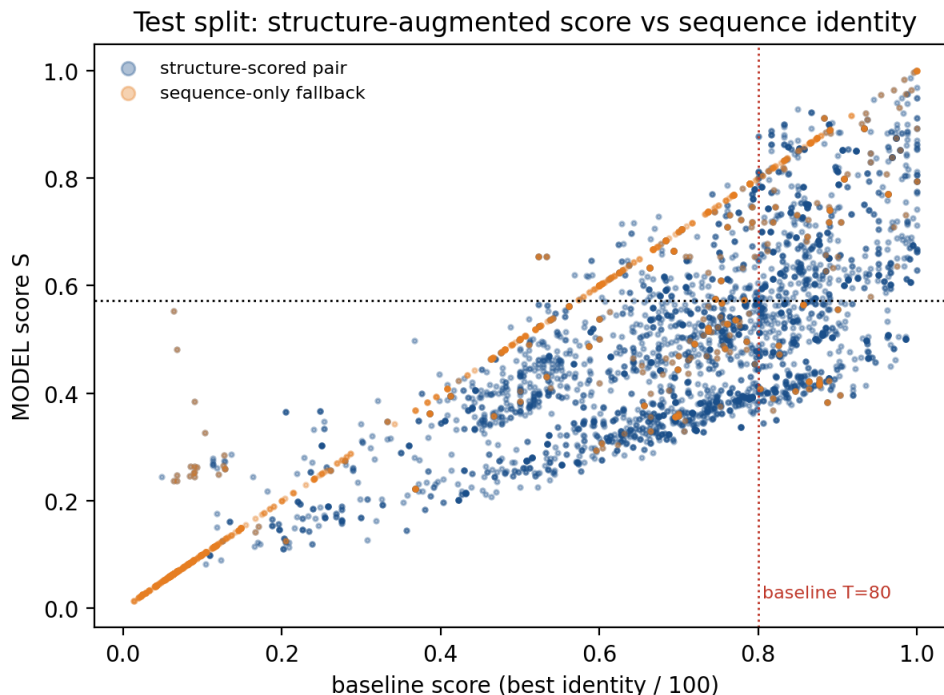


Figure 4: Model score versus baseline identity on the test split. Structure re-ranks pairs in both directions around the baseline’s  $T=80$  line; sequence-only fallback pairs (orange) lie on the diagonal by construction.

tiveneno Micrurus; Antivenin Micrurus fulvius; Antivenom B multicinctus N n naja; Banded krait antivenin QSMI; Bungarus Antivenom; Bungarus multicinctus antivenin; CSL Black Snake antivenom; CSL Polyvalent antivenom; CSL Taipan antivenom; CSL Tiger Snake antivenom; Habu antivenom; Haemato polyvalent snake antivenom QSMI; Malayan pit viper antivenin QSMI; Mamushi Antivenom Kaketsuken; Naja naja atra antivenom; Neuro polyvalent snake antivenom QSMI; Polyvalent Anti Snake Venom; Polyvalent Antisnake Venom; Polyvalent Snake Antivenin Asia; Polyvalent Snake Antivenom; Polyvalent Snake Antivenom Asia; Snake Venom Antiserum IP Asia; Snake antivenin IP Asia.

#### 4.8 Honest negatives

- **India addendum unsupported (gate M9).** Four CRITICAL/HIGH species occur in India per the committed gap table (*Bungarus magnimaculatus*, *Hypnale hypnale*, *Trimeresurus gramineus*, *T. malabaricus*); *none* have toxin records, below the locked minimum of three toxin-bearing species, so no India-specific scoring was computed or claimed. This directly contradicts the stale spec’s “India 4/3/240” numbers, which remain unverifiable and unused.
- **Structure fallbacks.** 28.7% of scored (query, product) rows rely wholly or partly on the sequence-only fallback (10,180 no-structure pairs; 3,812 near-zero alignments); 6 pairs had no both-exposed aligned positions ( $E=0$ , flagged). All are flagged in the committed score table.
- **Distinct-lineage toxins defeat both methods.** *Atractaspis* sarafotoxins (e.g. *A. engaddensis*, 3 toxins) share family labels with elapid 3FTx but no alignable homolog among covered species: baseline identity 0.0, model score 0.0, both correctly silent. Cross-reactivity inference cannot span genuinely

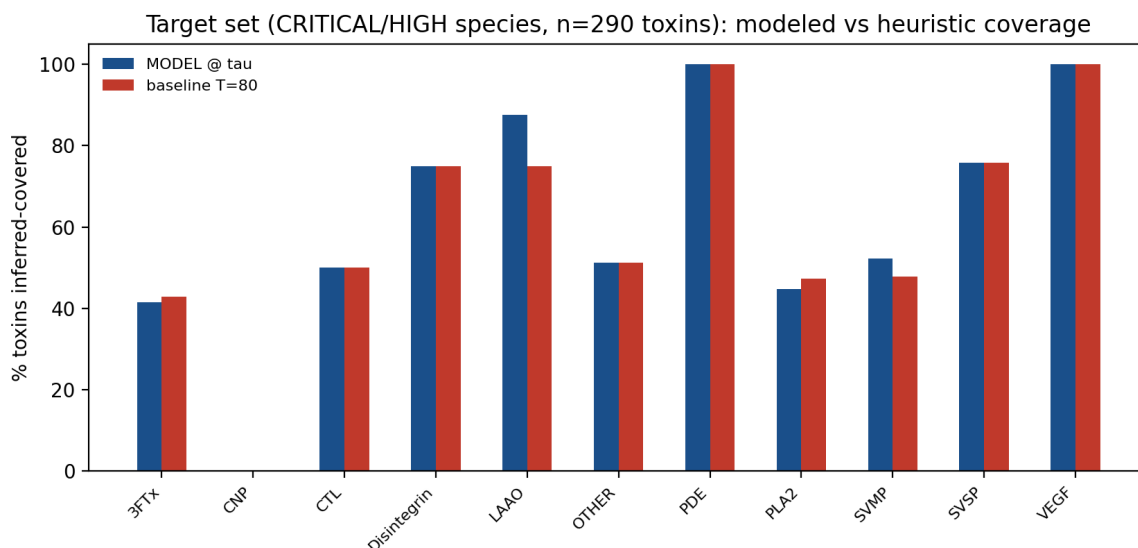


Figure 5: Target-set coverage by family: model at frozen  $\tau$  versus the committed heuristic at T=80.

independent toxin folds.

- **Small families are underpowered.** CNP and PDE contribute 2 target toxins each; per-family numbers there are descriptive only.
- **AUROC inversion vs the product prior** is reported above (Section 4.3) rather than suppressed.

## 4.9 Discordance analysis

Of the 290 target toxins, 39 (13.4%) flip verdict between the two methods: the model upgrades toxins whose structural match is stronger than raw identity suggests and downgrades identity-driven hits whose surfaces diverge. The full list follows; per-family net effects are small in both directions (Figure 5), so the model re-ranks rather than inflates coverage.

Table 3: Discordant target toxins: model at  $\tau$  versus committed heuristic at T=80 disagree on 39 of 290 toxins (13.4%). S = model max score; I = baseline best identity.

| accession | species                        | family | S      | I    | verdict       |
|-----------|--------------------------------|--------|--------|------|---------------|
| F8QN53    | <i>Vipera renardi</i>          | PLA2   | 0.6017 | 79.0 | model-only    |
| P0DI89    | <i>Bothrops leucurus</i>       | LAAO   | 0.579  | 70.3 | model-only    |
| P82896    | <i>Trimeresurus stejnegeri</i> | PLA2   | 0.7273 | 69.9 | model-only    |
| P84787    | <i>Protobothrops elegans</i>   | SVSP   | 0.549  | 86.8 | baseline-only |
| P84907    | <i>Bothrops leucurus</i>       | SVMP   | 0.4509 | 81.5 | baseline-only |
| P0DJ62    | <i>Bothrops leucurus</i>       | PLA2   | 0.6127 | 55.0 | model-only    |
| P0DKR3    | <i>Pseudocerastes fieldi</i>   | 3FTx   | 0.5639 | 87.7 | baseline-only |
| P0DM39    | <i>Trimeresurus albolabris</i> | OTHER  | 0.6639 | 77.7 | model-only    |
| P0DMU0    | <i>Protobothrops jerdonii</i>  | SVSP   | 0.762  | 76.2 | model-only    |

| accession  | species                        | family | S      | I    | verdict       |
|------------|--------------------------------|--------|--------|------|---------------|
| P20897     | <i>Crotalus ruber</i>          | SVMP   | 0.768  | 76.8 | model-only    |
| P84788     | <i>Protobothrops elegans</i>   | SVSP   | 0.5197 | 85.2 | baseline-only |
| Q6H3C9     | <i>Trimeresurus stejnegeri</i> | PLA2   | 0.7591 | 79.0 | model-only    |
| Q6H3D3     | <i>Trimeresurus stejnegeri</i> | PLA2   | 0.5398 | 80.4 | baseline-only |
| Q6H3D4     | <i>Trimeresurus stejnegeri</i> | PLA2   | 0.545  | 81.2 | baseline-only |
| Q7T1R0     | <i>Bungarus flaviceps</i>      | 3FTx   | 0.5994 | 77.6 | model-only    |
| Q98996     | <i>Daboia palaestinae</i>      | PLA2   | 0.5548 | 87.0 | baseline-only |
| A0A193CHJ3 | <i>Crotalus tzabcan</i>        | OTHER  | 0.5666 | 88.4 | baseline-only |
| D8MIA2     | <i>Bitis rhinoceros</i>        | SVSP   | 0.752  | 77.9 | model-only    |
| P24541     | <i>Eristocophis macmahoni</i>  | SVSP   | 0.3971 | 88.2 | baseline-only |
| P60623     | <i>Trimeresurus stejnegeri</i> | 3FTx   | 0.4471 | 86.2 | baseline-only |
| P81114     | <i>Trimeresurus albolabris</i> | OTHER  | 0.658  | 65.8 | model-only    |
| P81115     | <i>Trimeresurus albolabris</i> | OTHER  | 0.5977 | 76.9 | model-only    |
| Q3HTN1     | <i>Trimeresurus stejnegeri</i> | SVMP   | 0.739  | 77.4 | model-only    |
| Q3HTN2     | <i>Trimeresurus stejnegeri</i> | SVMP   | 0.732  | 77.0 | model-only    |
| Q71QI6     | <i>Trimeresurus stejnegeri</i> | SVSP   | 0.8288 | 77.4 | model-only    |
| Q71QI9     | <i>Trimeresurus stejnegeri</i> | SVSP   | 0.8033 | 78.2 | model-only    |
| Q71RQ7     | <i>Trimeresurus stejnegeri</i> | OTHER  | 0.5656 | 80.1 | baseline-only |
| Q8AY78     | <i>Trimeresurus stejnegeri</i> | SVSP   | 0.7615 | 79.5 | model-only    |
| Q9PWR6     | <i>Daboia palaestinae</i>      | PLA2   | 0.5497 | 85.5 | baseline-only |
| Q9YGJ7     | <i>Daboia palaestinae</i>      | PLA2   | 0.4722 | 82.6 | baseline-only |
| A0A2P1BSU3 | <i>Micrurus mipartitus</i>     | 3FTx   | 0.5844 | 57.1 | model-only    |
| A0AAU6NT02 | <i>Micrurus mipartitus</i>     | SVSP   | 0.5316 | 80.6 | baseline-only |
| A0AAU6NT91 | <i>Micrurus mipartitus</i>     | SVSP   | 0.5561 | 80.5 | baseline-only |
| B7FDI0     | <i>Vipera nikolskii</i>        | 3FTx   | 0.5596 | 92.1 | baseline-only |
| P34075     | <i>Naja annulata</i>           | 3FTx   | 0.5855 | 77.0 | model-only    |
| Q71RQ9     | <i>Trimeresurus stejnegeri</i> | OTHER  | 0.6203 | 70.5 | model-only    |
| Q71RR0     | <i>Trimeresurus stejnegeri</i> | OTHER  | 0.4036 | 82.2 | baseline-only |
| A0A077LA68 | <i>Protobothrops elegans</i>   | OTHER  | 0.487  | 94.6 | baseline-only |
| D5J9P3     | <i>Bungarus flaviceps</i>      | 3FTx   | 0.571  | 80.2 | baseline-only |

## 5 The tool

modeling/code/coverage\_model.py answers per-species and per-toxin coverage queries against the committed artifacts: given a species it prints its gap class, its panel toxins, and each toxin's modeled score, covering products at  $\tau$ , and the baseline comparison; given a toxin accession it prints the same card. Example (*Atractaspis engaddensis*, HIGH gap class): 3 toxins, 0/3 model-covered, agreeing with the baseline – an honest negative rather than a forced answer. The tool is deliberately read-only over the committed tables; re-running the pipeline regenerates every number it prints.

### 5.1 Per-species detail (test split)

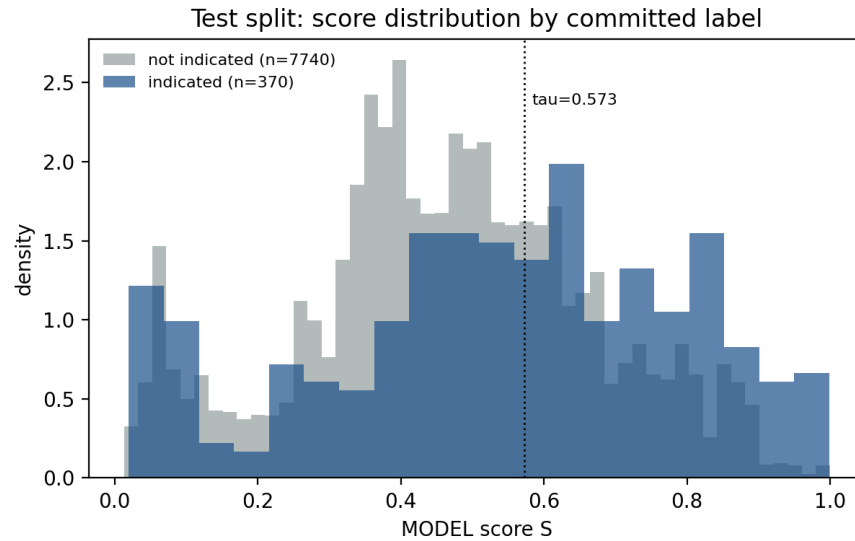


Figure 6: Test-split score distributions by committed label. The classes separate but overlap – the honest picture behind the AUPRC numbers;  $\tau$  sits at the calibration optimum, not at a visual valley.

Table 4: Held-out test species: panel size, true indicated products, product recovery and inventory coverage at frozen  $\tau = 0.5727$ . One of the 21 test species had no scorable (toxin, product) pair (no indicated reference passed the prefilter) and is omitted here – recorded as a negative.

| species                         | toxins | products | recovery | inv. covered |
|---------------------------------|--------|----------|----------|--------------|
| <i>Acanthophis spp</i>          | 15     | 2        | 0.50     | 0.47         |
| <i>Bitis gabonica</i>           | 9      | 2        | 0.50     | 0.44         |
| <i>Bothriechis schlegelii</i>   | 3      | 1        | 1.00     | 1.00         |
| <i>Bothrops mattogrossensis</i> | 5      | 2        | 0.00     | 0.40         |
| <i>Bungarus fasciatus</i>       | 13     | 3        | 0.33     | 0.54         |
| <i>Crotalus atrox</i>           | 44     | 2        | 1.00     | 0.75         |
| <i>Crotalus horridus</i>        | 45     | 1        | 1.00     | 0.89         |
| <i>Crotalus molossus</i>        | 9      | 2        | 1.00     | 0.78         |
| <i>Crotalus viridis</i>         | 20     | 2        | 1.00     | 0.70         |
| <i>Daboia russelii</i>          | 40     | 4        | 1.00     | 0.78         |
| <i>Dendroaspis jamesoni</i>     | 9      | 2        | 1.00     | 0.56         |
| <i>Gloydius halys</i>           | 28     | 1        | 0.00     | 0.61         |
| <i>Gloydius ussuriensis</i>     | 12     | 1        | 1.00     | 0.92         |
| <i>Naja annulifera</i>          | 18     | 1        | 1.00     | 0.94         |
| <i>Naja melanoleuca</i>         | 8      | 2        | 1.00     | 0.75         |
| <i>Naja samarensis</i>          | 1      | 1        | 1.00     | 1.00         |
| <i>Naja sputatrix</i>           | 31     | 1        | 0.00     | 0.81         |
| <i>Ophiophagus hannah</i>       | 118    | 2        | 0.50     | 0.25         |
| <i>Porthidium ophryomegas</i>   | 1      | 1        | 0.00     | 0.00         |

| species               | toxins | products | recovery | inv. covered |
|-----------------------|--------|----------|----------|--------------|
| <i>Vipera ursinii</i> | 1      | 1        | 0.00     | 1.00         |

## 5.2 Per-species detail (target set)

Table 5: Target set: all 38 CRITICAL/HIGH species with toxin records. Toxins = in-panel toxins; model / baseline = toxins inferred-covered by the cascade at  $\tau$  and by the committed heuristic at  $T=80$ .

| species                               | gap class | toxins | model | baseline |
|---------------------------------------|-----------|--------|-------|----------|
| <i>Atheris chlorechis</i>             | HIGH      | 1      | 1     | 1        |
| <i>Atheris squamigera</i>             | HIGH      | 1      | 1     | 1        |
| <i>Atractaspis andersonii</i>         | CRITICAL  | 2      | 0     | 0        |
| <i>Atractaspis bibronii</i>           | HIGH      | 1      | 0     | 0        |
| <i>Atractaspis engaddensis</i>        | HIGH      | 3      | 0     | 0        |
| <i>Atractaspis irregularis</i>        | HIGH      | 2      | 0     | 0        |
| <i>Atropoides spp</i>                 | HIGH      | 2      | 2     | 2        |
| <i>Bitis nasicornis</i>               | HIGH      | 3      | 1     | 1        |
| <i>Bitis rhinoceros</i>               | HIGH      | 10     | 3     | 2        |
| <i>Bothriechis spp</i>                | HIGH      | 1      | 1     | 1        |
| <i>Bothrops bilineatus</i>            | CRITICAL  | 1      | 1     | 1        |
| <i>Bothrops leucurus</i>              | CRITICAL  | 10     | 6     | 5        |
| <i>Bungarus flaviceps</i>             | HIGH      | 22     | 6     | 6        |
| <i>Calliophis bivirgatus</i>          | HIGH      | 22     | 11    | 11       |
| <i>Cerrophidion sasai</i>             | HIGH      | 1      | 1     | 1        |
| <i>Crotalus ruber</i>                 | HIGH      | 3      | 2     | 1        |
| <i>Crotalus tzabcan</i>               | HIGH      | 8      | 7     | 8        |
| <i>Daboia palaestinae</i>             | CRITICAL  | 7      | 2     | 5        |
| <i>Eristocophis macmahoni</i>         | HIGH      | 6      | 1     | 2        |
| <i>Gloydus tsushimaensis</i>          | HIGH      | 1      | 1     | 1        |
| <i>Micrurus lemniscatus</i>           | HIGH      | 1      | 0     | 0        |
| <i>Micrurus mipartitus</i>            | HIGH      | 15     | 7     | 8        |
| <i>Micrurus surinamensis</i>          | HIGH      | 6      | 0     | 0        |
| <i>Naja anchietae</i>                 | CRITICAL  | 1      | 1     | 1        |
| <i>Naja annulata</i>                  | HIGH      | 2      | 2     | 1        |
| <i>Naja christyi</i>                  | HIGH      | 1      | 1     | 1        |
| <i>Naja pallida</i>                   | HIGH      | 3      | 3     | 3        |
| <i>Naja sagittifera</i>               | CRITICAL  | 2      | 1     | 1        |
| <i>Proatheris superciliaris</i>       | HIGH      | 2      | 0     | 0        |
| <i>Protobothrops elegans</i>          | HIGH      | 23     | 12    | 15       |
| <i>Protobothrops jerdonii</i>         | HIGH      | 19     | 18    | 17       |
| <i>Protobothrops mangshanensis</i>    | HIGH      | 2      | 1     | 1        |
| <i>Pseudocerastes fieldi</i>          | HIGH      | 3      | 1     | 2        |
| <i>Trimeresurus albolabris</i>        | HIGH      | 18     | 12    | 9        |
| <i>Trimeresurus purpureomaculatus</i> | HIGH      | 5      | 1     | 1        |

| species                        | gap class | toxins | model | baseline |
|--------------------------------|-----------|--------|-------|----------|
| <i>Trimeresurus stejnegeri</i> | HIGH      | 68     | 47    | 44       |
| <i>Vipera nikolskii</i>        | HIGH      | 5      | 4     | 5        |
| <i>Vipera renardi</i>          | HIGH      | 7      | 4     | 3        |

Table 6: Pipeline statistics (all artifacts committed; every count regenerable from `modeling/code/`).

| quantity                                                             | value          |
|----------------------------------------------------------------------|----------------|
| panel queries (all in-scope toxins with sequences)                   | 3,644          |
| of which target (CRITICAL/HIGH)                                      | 290            |
| scored (toxin, product) rows                                         | 62475          |
| unique shortlisted (toxin, reference) pairs                          | 78468          |
| structure-scored (status ok)                                         | 64470          |
| sequence-only fallback (no structure)                                | 10180          |
| fallback (alignment < 8 positions)                                   | 3812           |
| AFDB v6 models fetched (100%    models with usable C $\alpha$ trace) | 3,059          |
| calibration / test species                                           | 85 / 21        |
| calibration / test (toxin, product) pairs                            | 48,317 / 8,110 |

## 6 Discussion and limitations

The result is a real but modest win, stated precisely: under a species-level holdout that mirrors the actual use case (a new or uncovered species arrives; which existing products plausibly cover its toxins?), structural re-scoring of sequence shortlists improves pair ranking over the established heuristic, with a bootstrap CI that excludes zero under the pre-registered criterion. On the target set the two methods largely agree, which is itself the scientifically useful finding: the heuristic’s headline coverage estimate (about half the gap inventory plausibly covered) survives scrutiny by a stronger method, while the model re-ranks *which* products to trust per toxin.

Limitations bound every claim. The labels are manufacturer/regulatory *indications*, not binding measurements; pair-level precision is capped by label coarseness. AlphaFold models are predictions, and 261 panel accessions have none. The epitope term is a solvent-exposure and chemistry-class proxy, not a paratope–epitope simulation; polyclonal avidity, glycosylation, and neutralization (as opposed to binding) are all outside the model. The weight triple was fixed a priori for honesty, not optimized; a fitted model would likely score higher and prove less. And per the access-only gate, nothing here spent money, created accounts, or communicated as anyone.

## 7 Reproducibility

Every number in this paper regenerates from `modeling/code/` in eight steps: `panel.py` (panel + seeded split) → `baseline_check.py` (baseline equivalence) → `shortlist.py` (sequence stage) → `fetch_afdb.py` (structure acquisition) → `features.py` (structural features) → `score.py` (assembly, calibration, evaluation) → `target_coverage.py` (target set) → `figures.py`. Each step reads only committed inputs or earlier committed outputs against the committed B09 inputs and the sha256-locked AFDB fetch manifest. The gate file, split assignment, fetch manifest, score tables, and this paper’s sources are byte-locked in

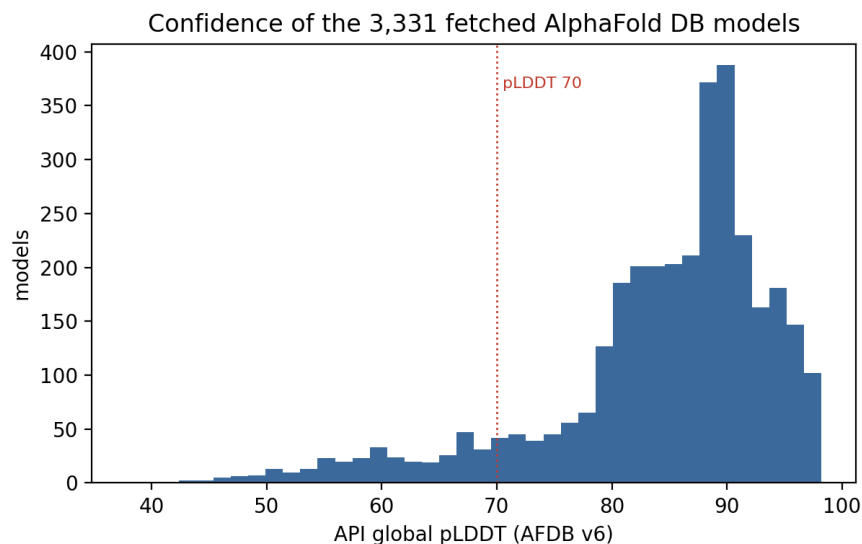


Figure 7: Structure confidence across the fetched model set: mean global pLDDT 84.3, 89.9% of models at  $\text{pLDDT} \geq 70$ , 0.8% below 50. Structural comparisons in this slice rest almost entirely on high-confidence models.

modeling/manifests/slice-manifest.md; the model PDB files (577 MB) are hash-locked and regenerable from their recorded URLs rather than committed, by size. Commit order – gates, design, results, seal – is the lock-order proof.

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